

Full-Stack Technology Audit & Maturity Report

GeoNexus Platform Architecture: Frontend, Backend, AI Multi-Agent & Database Systems

Doc Ref: GNX-AUDIT-2026-V1 | Scope: Complete Full-Stack Technology Audit & Readiness Guide

1. TECHNOLOGY MATURITY CLASSIFICATION FRAMEWORK

RED: BASIC (Level 1)	Foundational utilities, routine third-party packages, and standard dependencies used for basic functionality without custom architectural wrapping.
YELLOW: INTERMEDIATE (Level 2)	Standard industry frameworks, data manipulation libraries, and reliable infrastructure requiring structured configuration and pipelines.
GREEN: ADVANCED (Level 3)	High-complexity modules, strict end-to-end static type systems, 4-tier RBAC security, hybrid search algorithms, and automated test regression suites.
BLUE: MASTER (Level 4)	Elite architectural achievements: 8-agent autonomous DAG orchestration, zero-hallucination dual AI engine, polymorphic DB adapters, and bespoke CSS token system.

2. TECHNOLOGY DISTRIBUTION & PRODUCTION READINESS SCORECARD

Maturity Tier	Count / Ratio	Key Technologies & Representative Systems
BLUE (MASTER)	6 Components (20.0%)	Multi-Agent DAG, Dual AI Engine, Hybrid Search, Discrepancy Engine, Custom Token...
GREEN (ADVANCED)	16 Components (53.3%)	React 19, TypeScript 5.7, Vite, PyMuPDF, OCR Pipeline, 4-Tier RBAC, 158 Test Sui...
YELLOW (INTERMEDIATE)	5 Components (16.7%)	Flask REST API, Gunicorn WSGI, Chart.js, Pandas/OpenPyXL, FPDF2 Generator...
RED (BASIC)	3 Components (10.0%)	Lucide React Icons, Python-Dotenv, NPM Package Scripts...

EXECUTIVE AUDIT CONCLUSION: 73.3% ADVANCED & MASTER ARCHITECTURE

The platform demonstrates enterprise-grade maturity with zero legacy framework lock-in. Key strengths include an autonomous 8-agent DAG architecture, dual AI providers (cloud Gemini + zero-hallucination offline NLP), and a 158-test regression suite with 0 failures.

3. COMPREHENSIVE TECHNOLOGY INVENTORY MATRIX (30 COMPONENTS)

Domain Layer	Technology / Framework	Rank Tier	Architectural Role
Frontend UI Framework	React 19.0 (Hooks, Context, Strict)	GREEN - ADV	Declarative UI rendering & state syncing
Frontend Type Safety	TypeScript 5.7 (Strict Static Typing)	GREEN - ADV	End-to-end domain contracts & type guards
Frontend Build & HMR	Vite 6.1 (ESBuild, Rollup Bundling)	GREEN - ADV	Lightning-fast dev server & optimized bundles
Frontend Design System	Custom CSS3 Token Design System	BLUE - MASTER	100% bespoke dark tokens & 90% density
Frontend Data Visuals	Chart.js 4.4 & HTML5 Canvas	YELLOW - INT	Interactive mining telemetry & trends
Frontend Iconography	Lucide React 0.475 (SVG Icons)	RED - BASIC	Standardized vector icon visual cues
Frontend Responsive	Dual-Shell Architecture (Desktop/Mobile)	GREEN - ADV	Specialized desktop & mobile UI adapters
Frontend Testing	Vitest 3.0 & Testing Library	YELLOW - INT	Fast headless component & unit tests
Backend Runtime	Python 3.14 / 3.x Modern Runtime	GREEN - ADV	High-performance modular backend core
Backend Web Service	Flask 3.0+ REST API Application	YELLOW - INT	14 modular Blueprints & JSON endpoints
Backend WSGI Server	Gunicorn 23.0 WSGI Concurrency	YELLOW - INT	Production multi-worker HTTP serving
Backend Security	4-Tier RBAC Middleware	GREEN - ADV	ADMIN, OFFICER, AUDITOR, VIEWER security
Backend Architecture	Decoupled Domain Service Layer	GREEN - ADV	Clean separation of routes & business logic
AI Multi-Agent Core	Autonomous Multi-Agent DAG Framework	BLUE - MASTER	8 autonomous collaborating agents
AI Provider Engine	Polymorphic Dual-AI Provider Engine	BLUE - MASTER	Gemini API + Grounded Offline Engine
AI Discrepancy Engine	Cross-Document Conflict Engine	BLUE - MASTER	Multi-source fact validation & tolerances
AI Report Synthesis	Automated Regulatory Report Generator	GREEN - ADV	Synthesis of CIL statutory compliance docs
Search Engine	Hybrid Semantic + Lexical Retrieval	BLUE - MASTER	BM25 + TF-IDF vector reciprocal fusion
Vector Store	Custom Serialized Hybrid Vector Index	GREEN - ADV	Pickle-persisted vector store & scoring
Document Chunking	Recursive Semantic Chunking Pipeline	GREEN - ADV	Context & table-preserving text chunker
Document Ingestion	Multi-Format Parsing Engine	GREEN - ADV	PDF, DOCX, XLSX, CSV, TXT & image parsing
PDF Processing	PyMuPDF (Fitz) 1.28+ & PyPDF 4.0	GREEN - ADV	C-native high-speed text & layout extract
OCR Pipeline	Tesseract OCR & Image Preprocessing	GREEN - ADV	Binarization, deskewing & OCR confidence
Spreadsheet Engine	Pandas 2.2 & OpenPyXL 3.1	YELLOW - INT	Mining production dataframes & tables
Word Document Parser	Python-Docx 1.1	YELLOW - INT	DOCX XML parsing and paragraph extraction
PDF Generator	FPDF2 Programmatic PDF Generator	YELLOW - INT	Standardized statutory PDF report generation
Database Abstraction	Dual Polymorphic Database Adapters	BLUE - MASTER	Zero-change SQLite <-> PostgreSQL syncing
Database Schema	Normalized Relational Schema (12+ DBs)	GREEN - ADV	Strict foreign keys, constraints & indexing
PostgreSQL Driver	Psycopg3 Binary Driver	GREEN - ADV	Enterprise async Postgres connection pooling
Audit & Provenance	Immutable Audit Logging Engine	GREEN - ADV	Traceable user actions, IPs & agent logs

4. IN-DEPTH ARCHITECTURAL PROFILES: FRONTEND & BACKEND SERVICES

1. Custom CSS3 Token Design System

BLUE - MASTER

Source Path: frontend/src/styles/index.css

- Bespoke zero-dependency HSL design system eliminating UI framework overhead.

- Deep

2. React 19 & TypeScript 5.7 Architecture

GREEN - ADV

Source Path: frontend/src/

- Built on modern React 19 functional paradigm using stateful context providers (AuthContext, ToastContext).

- Strict

3. Dual-Shell Architecture (Desktop & Field Mobile)

GREEN - ADV

Source Path: frontend/src/components/layout/ & mobile/

- Adaptive runtime detection dynamically switching between multi-pane desktop shell and tactile mobile shell.

- Mobile

4. 4-Tier Role-Based Access Control (RBAC)

GREEN - ADV

Source Path: routes/auth_middleware.py

- Enforces 4 operational tiers: ADMIN (Full access), OFFICER (Operations), AUDITOR (Provenance), VIEWER (Public).

- Corre

5. Decoupled Modular Service Layer

GREEN - ADV

Source Path: services/ & routes/

- 14 isolated Flask Blueprints delegating all business logic to 10+ standalone domain services.

- Clean

5. IN-DEPTH ARCHITECTURAL PROFILES: AI, RETRIEVAL & PERSISTENCE

1. Autonomous Multi-Agent DAG Orchestration

BLUE - MASTER

Source Path: agents/ & agents/manager_agent.py

- 8 autonomous domain agents collaborating via Directed Acyclic Graph (DAG) workflows.

- Specifi

2. Polymorphic Dual-AI Provider Engine

BLUE - MASTER

Source Path: services/ai_provider.py

- Zero-downtime hot-swappable AI provider interface supporting Google Gemini 2.5 API.

- Offline

3. Cross-Document Discrepancy & Conflict Engine

BLUE - MASTER

Source Path: services/validation_service.py

- Automated fact normalization aligning numerical production metrics across daily reports, monthly returns, and statutory audits.

- Toleranc

4. Hybrid Semantic + Lexical Retrieval Engine

BLUE - MASTER

Source Path: services/retrieval_service.py

- Dual-channel retrieval fusing BM25 keyword matching with TF-IDF cosine vector similarity.

- Recip

5. Dual Polymorphic Database Adapter Pattern

BLUE - MASTER

Source Path: database/db.py & database/adapters/

- Unified database interface seamlessly supporting both SQLite (local development) and PostgreSQL (production).

- Identifi

6. 158-Suite Regression Test Harness

GREEN - ADV

Source Path: tests/

- Comprehensive automated test suite validating RBAC, agents, OCR, vector scoring, and API routes.

- 100%